Theory note S2 — The graph Laplacian, algebraic connectivity λ_2, and the link range under partial-band jamming

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Purpose. S2 summarises a whole network by one number, the algebraic connectivity λ_2 [1]. This note proves the fact the simulator relies on ($\lambda_2 > 0$ iff connected), states what each verification check tests, shows that in this model a link is up exactly when it is shorter than a *link range*, and uses that to explain the two sweeps. Mathematics used: symmetric matrices, eigenvalues, quadratic forms.

1 From links to a matrix

Take N nodes. From S1, each pair (i, j) has a bit error rate $P_b(d_{ij}, J/S, \rho)$; the link is *up* if $P_b \le \mathrm{BER}_{\mathrm{out}}$. Collect the result in the adjacency matrix

$$A_{ij} = \begin{cases} 1 & \text{link } (i,j) \text{ up, } i \neq j \\ 0 & \text{otherwise,} \end{cases}$$

symmetric with zero diagonal. The degree $d_i = \sum_j A_{ij}$ counts a node's live links; $D = \mathrm{diag}(d_1, \ldots, d_N)$. The *graph Laplacian* is

$$L = D - A.$$

Its rows sum to zero by construction: $L\mathbf{1} = D\mathbf{1} - A\mathbf{1} = (d_i) - (d_i) = 0$.

2 Three facts about L

(a) L is symmetric, so its eigenvalues are real. D and A are symmetric. Hence L has N real eigenvalues, which we order $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_N$, and an orthonormal eigenbasis.

(b) The quadratic form is a sum of squared differences over edges. For any vector $\mathbf{x} \in \mathbb{R}^N$,

$$\mathbf{x}^{\mathsf{T}} L \mathbf{x} = \sum_i d_i x_i^2 - \sum_{i,j} A_{ij} x_i x_j = \sum_{\{i,j\} \in E} (x_i - x_j)^2 \;\ge\; 0. \tag{1}$$

To see the second equality, expand $\sum_{\{i,j\}\in E}(x_i - x_j)^2$: each edge contributes $x_i^2 + x_j^2$ (which sums to $\sum_i d_i x_i^2$ over all edges touching i) and $-2x_i x_j$ (which sums to $-\sum_{i,j} A_{ij} x_i x_j$ since each unordered edge appears twice in the double sum).

(c) $\lambda_1 = 0$ always. From (b), L is positive semidefinite, so all $\lambda_k \ge 0$; and $L\mathbf{1} = 0$ shows 0 is attained with eigenvector $\mathbf{1}$. Thus $\lambda_1 = 0$ carries no information about the graph, which is why the second eigenvalue is used.

3 Theorem: $\lambda_2 > 0$ if and only if the graph is connected

Theorem 1. *The multiplicity of the eigenvalue 0 of L equals the number of connected components of the graph. In particular $\lambda_2 > 0 \iff$ the graph is connected.*

Proof. Suppose $L\mathbf{x} = 0$. Then $\mathbf{x}^\mathsf{T} L\mathbf{x} = 0$, and by (1) every edge has $x_i = x_j$. Following edges, $\mathbf{x}$ is constant on each connected component. Conversely, any vector constant on components is in the null space (each row of L only mixes a node with its neighbours, which share its value). So the null space is spanned by the indicator vectors of the components, and its dimension is the number of components c. Hence $\lambda_1 = \cdots = \lambda_c = 0$ and $\lambda_{c+1} > 0$; with $c = 1$ this reads $\lambda_2 > 0$. □

`VERIFICATION 2` is a test of the implementation, not of the theorem: for every graph evaluated during the sweeps (87,000 evaluations; many are repeats, because the same placements are thresholded at different ranges) it computes both $[\lambda_2 > 10^{-9}]$ and a breadth-first search, and requires them to agree. Zero mismatches means the adjacency construction, the Laplacian, and the eigensolver are consistent with an independent algorithm. (The 10^{-9} tolerance absorbs floating-point round-off; λ_2 of a disconnected graph comes back as $\pm 10^{-15}$, not exactly 0.)

4 Two exact benchmarks

Proposition 1 (Complete graph)**.** *For K_N (every pair linked), $\lambda_2 = N$.*

Proof. $A = \mathbf{1}\mathbf{1}^\mathsf{T} - I$ and $D = (N-1)I$, so $L = NI - \mathbf{1}\mathbf{1}^\mathsf{T}$. On the subspace $\mathbf{x} \perp \mathbf{1}$ we have $\mathbf{1}\mathbf{1}^\mathsf{T}\mathbf{x} = 0$, so $L\mathbf{x} = N\mathbf{x}$: the eigenvalue N has multiplicity $N-1$. Together with $\lambda_1 = 0$ this is the whole spectrum, and $\lambda_2 = N$. □

`VERIFICATION 1` checks this for $N = 10, 20, 30$. It catches a wrong sign convention for L or a wrong eigenvalue index.

Degree bound. Fiedler [1, 3.5] showed that $\lambda_2 \le \frac{N}{N-1}\delta_{\min}$, where $\delta_{\min}$ is the smallest node degree; K_N attains it with equality. The bound is used here without proof, as a third consistency check on every graph (`VERIFICATION 3`).

5 Why λ_2 and not "number of live links"

The number of live links does not say where they are: losing the few links that join two clusters splits the network, losing the same number elsewhere does not. λ_2 reflects that structure. It also has a dynamical meaning: for the consensus dynamics $\dot{\mathbf{x}} = -L\mathbf{x}$ (each node moves toward the average of its neighbours), the disagreement decays at least as fast as $e^{-\lambda_2 t}$ [2, Corollary 1]. A falling λ_2 therefore means *slower* coordination before it means *no* coordination, which a binary connected/disconnected flag cannot show. For that reason the script records, besides the collapse threshold, the J/S at which the mean λ_2 has fallen to half its unjammed value (the "knee").

6 Links as a distance threshold

The link range. In this model the jammer is far from all nodes, so every receiver sees the same J, while the received signal falls with link length d. Both $E_b/N_0(d)$ and $E_b/N_J(d)$ decrease with d, so the link BER increases with d, and

$$P_b(d_{ij}) \le \mathrm{BER}_{\mathrm{out}} \iff d_{ij} \le r(J/S, \rho),$$

where the *link range* r solves $P_b(r) = \mathrm{BER}_{\mathrm{out}}$. The link graph is therefore a random geometric graph [3]: N uniform points, joined when closer than r. `VERIFICATION 5` confirms that the BER rule and the distance rule give identical adjacency matrices for every graph of the J/S sweep.

The jammer's choice of ρ. λ_2 is the minimum of $\mathbf{x}^{\mathsf{T}}L\mathbf{x}/\mathbf{x}^{\mathsf{T}}\mathbf{x}$ over $\mathbf{x} \perp \mathbf{1}$ (the Rayleigh quotient of a symmetric matrix, restricted to the vectors orthogonal to the first eigenvector). Removing edges cannot increase it: by (1), fewer edges make $\mathbf{x}^{\mathsf{T}}L\mathbf{x}$ smaller for every $\mathbf{x}$, hence also its minimum. A smaller range removes edges, so the ρ that minimises λ_2 is the ρ that minimises r. It is the same for every placement and every N, and the script computes it once per J/S value (`worst_rho()`). `VERIFICATION 4` checks the monotonicity placement by placement; `VERIFICATION 6` checks that the ρ with the lowest mean λ_2 on the grid of the ρ-sweep is the grid neighbour of the range-minimising ρ.

Which ρ that is (figure `s2_lambda2_vs_rho`). Minimising the range means bringing down the strongest possible link, which is the question answered in Section 4 of the S1 note:

$$\rho_{\mathrm{out}} \approx \frac{\mathrm{BER}_{\mathrm{out}}}{\mathrm{Q}(\sqrt{x^*})} = 8.55\,\mathrm{BER}_{\mathrm{out}} = 0.0086.$$

The exact search gives 0.0091 at $J/S_{\mathrm{ref}} = -14\,\mathrm{dB}$ (thermal noise moves it slightly) and λ_2 is lowest at the nearest grid point, 0.0089. This is an order of magnitude below the 0.135 of the S1 figure, but for a simple reason: both are $\rho^* = 0.709/(E_b/N_J)$, evaluated at $E_b/N_J = 8\,\mathrm{dB}$ in the S1 figure and at the outage point 19.2 dB here. It is not a network effect, and it scales with the assumed criterion ($\mathrm{BER}_{\mathrm{out}} = 10^{-2}$ would give $\rho_{\mathrm{out}} = 0.086$). The rise of λ_2 on the left of the figure has a separate cause: a jammed hop has a BER of at most $\frac{1}{2}$, so $P_b \le \rho/2$ plus the thermal term, and no jammer power can bring a link down once $\rho < 2\,\mathrm{BER}_{\mathrm{out}}$.

One-channel limit. $\rho_{\mathrm{out}} = 0.0086$ is just below $1/N_{\mathrm{ch}} = 0.01$, the smallest fraction a jammer can occupy with 100 channels; the figure shades that region. Enforcing $\rho \ge 0.01$ changes the range at $-14\,\mathrm{dB}$ from 538.8 m to 539.7 m and the thresholds below by 0.03 dB.

7 The range–power relation and the collapse threshold

Closed form. When jamming dominates thermal noise, the longest surviving link has $E_b/N_J = 0.0829/\mathrm{BER}_{\mathrm{out}}$ (S1 note, Section 4). With $E_b/N_J(d) = G_p - J/S_{\mathrm{ref}} - 10n\log_{10}(d/d_{\mathrm{ref}})$ in dB,

$$J/S_{\mathrm{ref}}(r)\,[\mathrm{dB}] \;\approx\; G_p - 10\log_{10}\frac{0.0829}{\mathrm{BER}_{\mathrm{out}}} - 10\,n\log_{10}\frac{r}{d_{\mathrm{ref}}}. \tag{2}$$

Three consequences:

- every threshold moves by $+10\,\mathrm{dB}$ per tenfold increase of N_{ch} (as long as $\rho_{\mathrm{out}} \ge 1/N_{\mathrm{ch}}$), and by $-10\,\mathrm{dB}$ per tenfold *decrease* of $\mathrm{BER}_{\mathrm{out}}$;
- the *spacing* between the curves for different N is $10\,n\log_{10}(r_{N_1}/r_{N_2})$: independent of $\mathrm{BER}_{\mathrm{out}}$, G_p and d_{ref}, but proportional to the path-loss exponent n;
- the absolute values of J/S_{ref} carry no meaning beyond the chosen d_{ref}; the geometry enters only through r/L.

Collapse threshold (figure `s2_lambda2_vs_js`). J/S^* is defined as the J/S_{ref} at which half of the random placements are still connected. The script sweeps J/S_{ref} in 1 dB steps over 500 placements per N, interpolates the 0.5 crossing linearly, and obtains a 95 % interval by resampling the placements (bootstrap, 1000 resamples).

N	J/S^* [dB]	95 % interval	range at J/S^*	Eq. (2)	50 % knee
10	-11.79	-11.99 .. -11.61	$0.421\,L$	-11.68	-17.35
20	-9.42	-9.55 .. -9.29	$0.323\,L$	-9.36	-17.22
30	-7.92	-8.08 .. -7.80	$0.272\,L$	-7.88	-17.18

Going from 10 to 20 nodes buys 2.4 dB of jamming tolerance and from 20 to 30 another 1.5 dB: the gain per added node shrinks, because the range needed to keep N random points connected shrinks more slowly as N grows. Eq. (2) reproduces the simulated thresholds within 0.11 dB (`VERIFICATION 7`); the remainder is thermal noise. The knee, in contrast, is almost the same for all three sizes (-17.2 to -17.4 dB).

Limits of the model. Both results rest on the hard outage threshold without coding and on a distant jammer; the README lists these assumptions and what would change without them.

8 Correspondence with the code

Mathematics	Code
$A_{ij} = [P_b \le \mathrm{BER}_{\mathrm{out}}]$	`adjacency()`
$L = D - A$	`lap = np.diag(a.sum(axis=1)) - a`
λ_2 (2nd smallest, symmetric)	`np.linalg.eigvalsh(lap)[1]`
connected? (independent)	`is_connected_bfs(a)`
link range $r(J/S, \rho)$	`link_range()`
range-minimising ρ	`worst_rho()`
Eq. (2)	`js_ref_closed_form()`
0.5 crossing, knee	`crossing()`
K_N: $\lambda_2 = N$	`VERIFICATION 1`
$\lambda_2 > 0 \iff$ connected	`VERIFICATION 2`
$\lambda_2 \le \frac{N}{N-1}\delta_{\min}$	`VERIFICATION 3`
λ_2 non-increasing in J/S	`VERIFICATION 4`
BER rule = distance rule	`VERIFICATION 5`
λ_2-minimising ρ = range-minimising ρ	`VERIFICATION 6`
Eq. (2) vs simulation	`VERIFICATION 7`

References

[1] M. Fiedler, "Algebraic connectivity of graphs," *Czechoslovak Mathematical Journal*, vol. 23, no. 2, pp. 298–305, 1973.

[2] R. Olfati-Saber, J. A. Fax, and R. M. Murray, "Consensus and cooperation in networked multi-agent systems," *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, Jan. 2007.

[3] M. Penrose, *Random Geometric Graphs*. Oxford University Press, 2003.